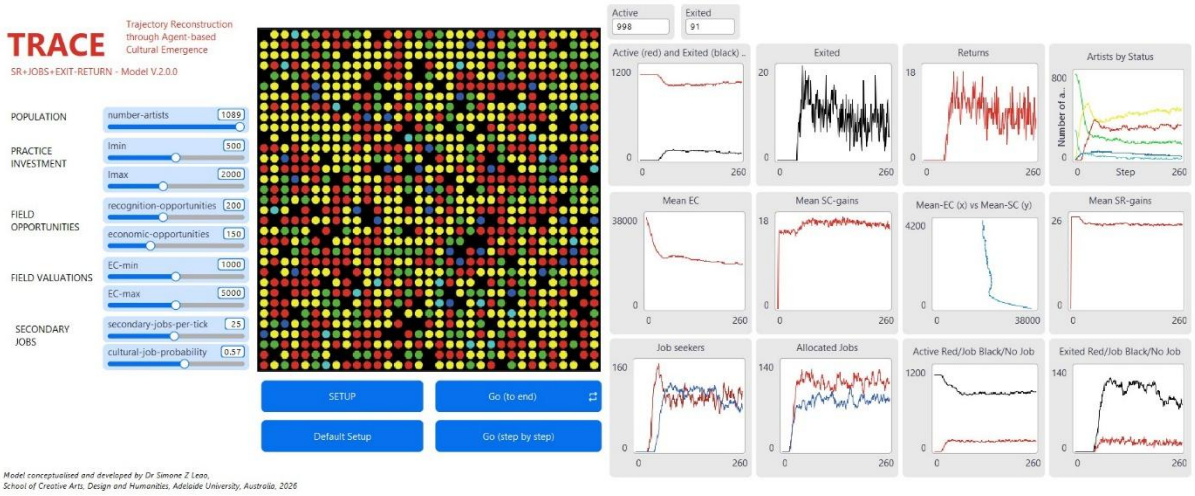
TRACE-v2

Trajectory Reconstruction through Agent-based Cultural Emergence (SR+Jobs+Exit+Return) VERSION 2



Model	TRACE - Trajectory Reconstruction through Agent-based Cultural Emergence
Type	Agent-based model
Version	2.0 (SR+Jobs+Exit+Return)
Year	2026
Platform	Netlogo 7.0.4
Author	Dr Simone Zarpelon Leao
Purpose	Research and academic use
License	CC BY-NC 4.0

1. Overview

TRACE-v2 (*Trajectory Reconstruction through Agent-based Cultural Emergence*) is an agent-based model in which heterogeneous artists respond to changing economic circumstances and employment opportunities while allocating time between artistic practice and secondary work. By examining career persistence, multiple job-holding, exit and re-entry, the model investigates how economic viability, recognition and self-realisation may follow different paths rather than necessarily reinforcing one another.

The TRACE-v2 model, code, documentation, and associated resources are publicly available through the GitHub repository, via Zenodo DOI.

2. Research Question

TRACE-v2 model asks: How do economic resources, symbolic recognition, secondary employment and self-realisation interact over time to shape the sustainability and trajectories of artistic careers?

3. Theoretical foundation

TRACE-v2 integrates selected theoretical propositions from three complementary perspectives: Bourdieu's account of the cultural field informs the representation of capital and symbolic recognition; Menger's analysis of artistic labour markets informs the treatment of occupational uncertainty and multiple job-holding; and Throsby's work-preference model informs the representation of intrinsic returns and the allocation of time between artistic and non-artistic work. These perspectives are brought together in an exploratory computational laboratory that generates artificial career trajectories through interacting mechanisms and stochastic events. Rather than reproducing actual career histories or forecasting individual outcomes, the model makes theoretical assumptions explicit and examines how different career patterns may emerge under a shared set of rules. It supports theoretical exploration and empirical inquiry rather than establishing independent causal effects.

4. Model description

TRACE follows an agent-based modelling approach in which individual artists interact within a competitive cultural field to generate emergent career trajectories. The model is structured according to the ODD (Overview, Design concepts, Details) protocol and represents the dynamic relationship between artistic investment, work tome allocation, competition for opportunities, multiple jobs-holding, symbolic recognition, economic returns, self-realisation, and career persistence, withdrawal, and re-entry.

4.1 Agents and environment

The model represents artists as autonomous agents operating within a closed artistic field. Each artist is characterised by economic capital (EC), symbolic capital (SC), self-realisation (SR), an investment disposition and a self-realisation propensity parameters representing differences in practice-oriented behaviour, and state variables indicating whether the artist remains active or has exited the field, and whether the artist is seeking for a secondary job. The environment represents a competitive cultural field with a fixed number of recognition and economic opportunities available at each simulation step. A fixed number of secondary jobs are available at each simulation step, a certain proportion of them are cultural related jobs.

4.2 Processes:

At each simulation step, every active artist progresses through six sequential processes:

a. **investment in artistic practice:** active artists first invest economic resources into artistic practice. The investment depends on their disposition to invest, minimum required investment, and maximum allowed investment. It is also influenced by an investment scaling parameter which reduces the investment if the artist economic capital is low.
b. **allocation of recognition opportunities and symbolic gains:** the field makes a certain number of recognition opportunities available to artists at every step. They are scarce. Artists compete for those through a weighted random selection process. The weighting is based on the symbolic capital profile of the field. Artists with higher symbolic capital are more likely but are not guaranteed to be selected. After artists are selected, the opportunities are valued through a bounded randomised process. These values are accumulated to the symbolic capital of the respective artists.
c. **allocation of economic opportunities and economic gains:** the field makes a certain number of economic opportunities available to artists at every step. They are scarce. Artists compete for those through a weighted random selection process. The weighting is based on the symbolic capital profile of the field. Artists with higher symbolic capital are more likely but are not guaranteed to be selected. After artists are selected, the opportunities are valued through a bounded randomised process. These values are added to the economic capital of the respective artists. Note that steps b and c are independent, but both allocate opportunities considering the symbolic capital of artists in the field.
d. **evaluation of job-seeking need:** artists compare their current economic capital to the required minimum investment in their artistic practice for the next 8 steps. If the capital is not enough, they become jobseekers, and start competing for the scarce available jobs.
e. **update working hours during multiple job-holding:** If an artist gets a secondary job alongside their artistic practice, they are informed of the job's working hours. duration and hourly pay rate. The artistic works are reduced accordingly. Artists' self-realisation declines accordingly, as it is a function of artistic hours of work and the self-realisation propensity.
f. **evaluation of continuation in or exit from the cultural field:** If artists fail to get a secondary job and their EC declines to a level below the required investment for the current step, they exit the cultural field. The exit is not necessarily permanent. While in exit they do not compete for recognition and economic opportunities, but they remaining seeking for secondary jobs. If they get a job and accumulate enough EC to cover the investment in artistic practice, they re-enter the cultural field.

4.3 Model default parameters

The default model consists of 1,089 artists simulated over 260-time steps (each step is equivalent to a week). All artists begin with an initial economic capital randomly chosen between $23,200 and $50,000, a symbolic capital of 0, and a self-realisation of 0. Investment disposition values are randomly assigned from a range of 0.20 to 0.60, with investment amounts constrained by minimum and maximum investment values ($500 to $2,000), and an investment scaling parameter (K=$20,000). Self-realisation propensity values are randomly assigned from a range of 0.20 to 1. 200 Recognition and 150 economic opportunities are provided by the field at each simulation step. Each recognition opportunity is valued by the field between a minimum and maximum values (40 and 120 units). Each economic opportunity is valued by the field between a minimum and maximum values ($1,000 and $5,000). Artists work 42 hours/week. 25 secondary jobs are provided in the field at each simulation step, of which 57% are cultural jobs and 43% are non-cultural. The characteristics of each job is randomly selected based on: hours of work (between 5-42 hours/week), payment rate (between $50-$120/hours), and duration (between 2-26 weeks). The default configuration represents a closed field in which artists with comparable economic and symbolic endowments compete for scarce opportunities, while career outcomes emerge from interactions between individual dispositions and field-level conditions.

5. Installation and requirements

5.1 Running TRACE in a desktop environment

TRACE requires the following software:

a. NetLogo (developed and tested using NetLogo 7.0.4)

b. Operating system: Windows, macOS, or Linux

No external NetLogo extensions or libraries are required to run the model.

To run TRACE, download and install NetLogo, open the **TRACE_model_v2.nlogox** model file.

Analysis of simulation outputs may require additional software packages (e.g., R or Python), depending on the analytical approach adopted.

5.2 Running TRACE in a web environment

An interactive online version of TRACE is available through NetLogo Web. To run this version this there is no need for software installation.

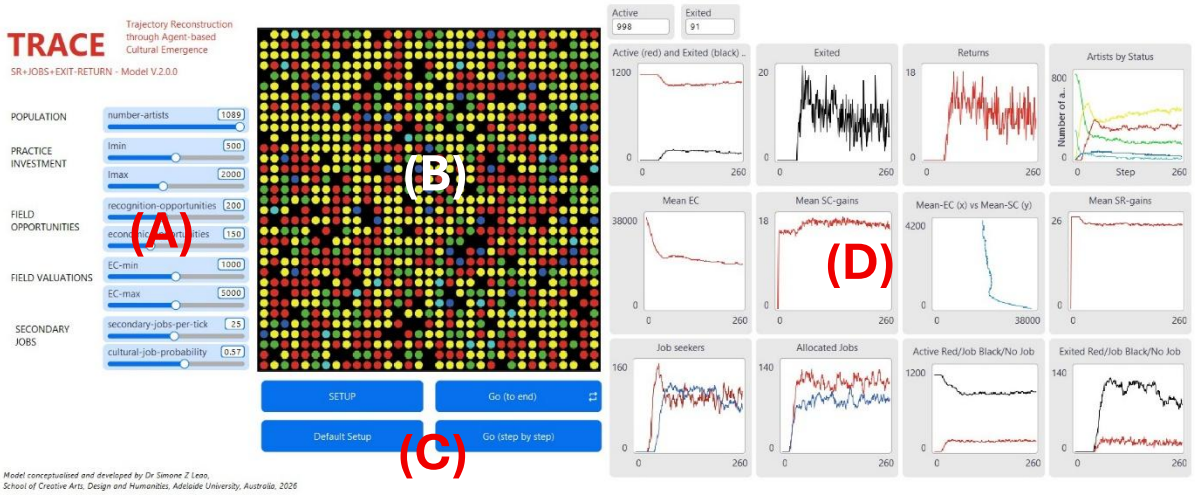
To run TRACE just click at

https://netlogoweb.org/launch#https://simonezleao.github.io/TRACE_ABM_model2/TRACE_model_v2_Web.nlogox

In this version, the user can run the model, visualise the simulation, and test the results if parameters are changed. However, the web environment does not allow for the outputs to be saved for future analysis.

6. Model Interface description

The model interface has four main parts: (a) parameters setting area; (b) simulation visualisation area; (c) processes buttons; and (d) results visualisation area. They are described in detail as follows.

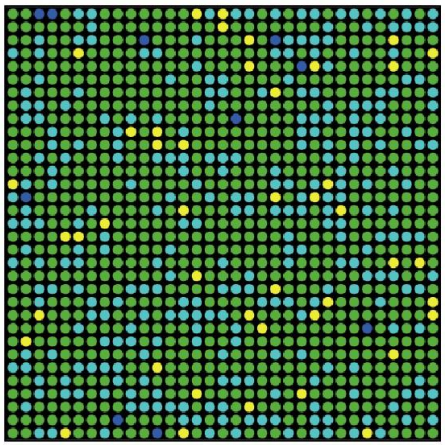


The parameters setting area (A) contains sliders for 9 model parameters. Through these sliders the user can change values for all or selected parameters within defined ranges. By changing parameters values a model user can test how the change affects the model simulation results.

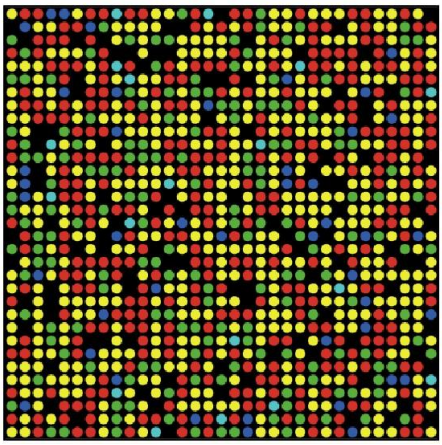


number-artists	Defines the total number of artists at the start of a simulation (max 1,089 artists)
lmin	Defines the minimum investment amount an artist needs to invest in practice ($1 to $1,000)
lmax	Defines the maximum investment amount an artist is allowed to invest in practice ($1 to $5,000)
recognition-opportunities	Defines the number of recognitions available in the field at each simulation step (0 to 500)
economic-opportunities	Defines the number of economic events available in the field at each simulation step (0 to 500)
EC-min	Defines the minimum valuation for an economic opportunity ($1 to $2,000)
EC-max	Defines the maximum valuation for an economic opportunity ($1 to $10,000)
secondary-jobs-per-tick	Defines the number of secondary jobs available in the field at each simulation step (0 to 50)
cultural-job-probability	Defines the % among secondary jobs which are of cultural type (0 to 100)

The simulation visualisation area (B) is the space representing the cultural field. It is a grid with 33 rows x 33 columns. Each cell can hold only one artist (therefore the maximum 1,089 initial artists). This is an abstract space which works as an infographic, not a geographical area.



1,089 artists at simulation step 0 with ECs between $23,200 and $50,000



Active artists still remaining in the field at the end of the simulation, with varied economic conditions

Artists are visualised in six colours: blue (high economic capital, EC≥$50,000); cyan (sustainable economic capital, EC≥$42,000 and EC<$50,000); green (average economic capital, EC≥$23,200 and EC<$42,000); yellow (precarious economic capital, EC≥$10,000 and EC<$23,200); red (critical economic capital, EC<$10,000); and black (exited from the field). The first five colours represent artists remaining active in the field.

The process buttons area (C) provides the activations for running the model. Four buttons are provided, two for setup and two for running the model, as described below.

This button sets up the model for the parameter values selected by the user. This is recommended for tests.

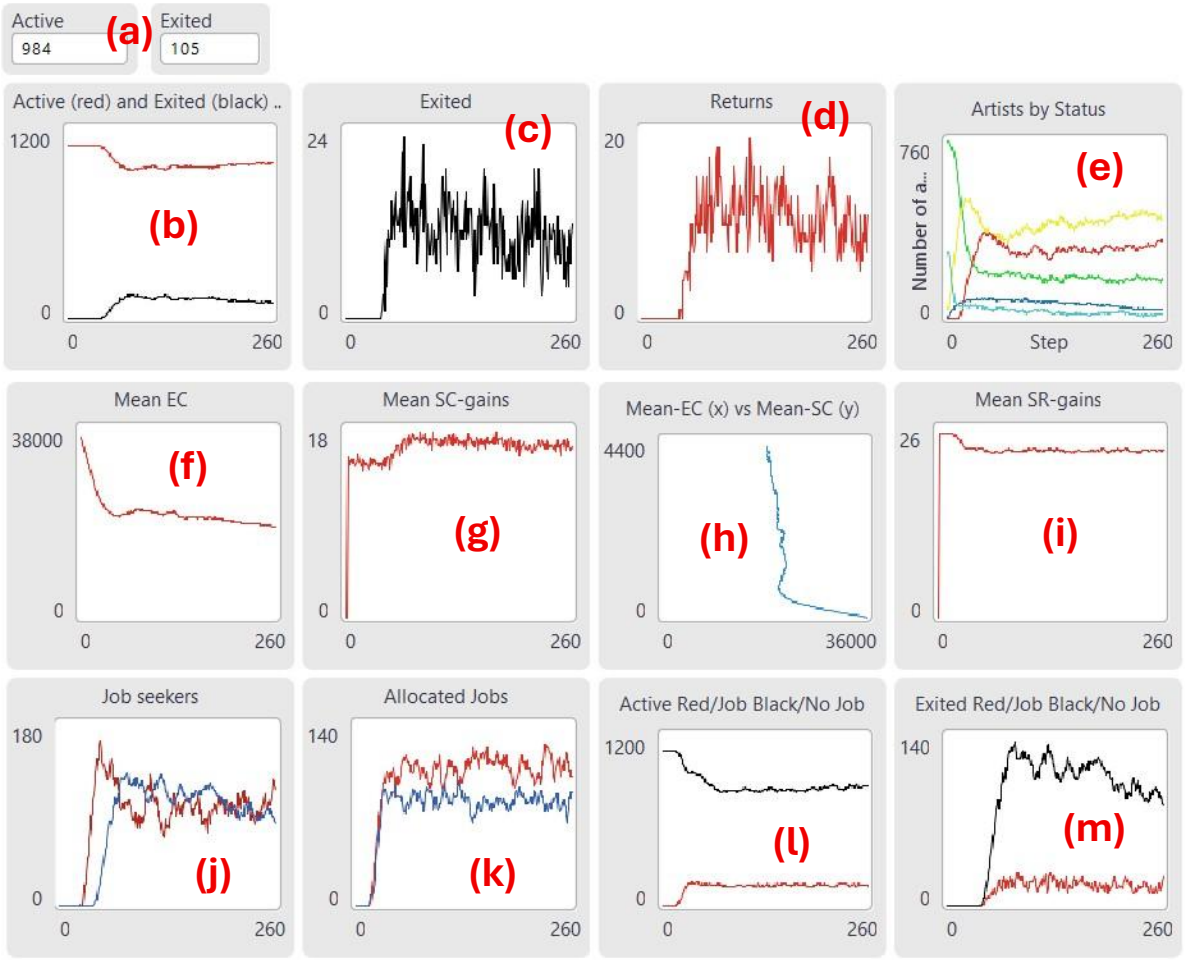
This button runs the model from start to the last simulation step continuously. Additional clicks in this button pauses or resumes the simulation. The speed of the run can be adjusted.



This button sets up the model for the default values of the model parameters

This button runs the model step by step. The user must click the button multiple times for a simulation. It is useful for visualisation of results mid simulation.

The results visualisation area (D) is a dashboard with several plots and monitors which present the results of the simulation as it happens. This way the user gets an insight of how the model parameters and outputs change and vary over time.



a. The monitors show the number of artists who are active and are exited at every simulation step. It is continuously updated to the latest step of the simulation along running.
b. The plot shows the number of active and exited artists along the simulation, storing the progress over time.
c. The plot shows the number of artists exited from the field at each step along the simulation.
d. The plot shows the number of artists returning to the field at each step along the simulation.
e. The plot shows the number of active artists along the simulation, divided economic capital condition, storing the progress over time. blue (high economic capital, EC≥$50,000); cyan (sustainable economic capital, EC≥$42,000 and EC<$50,000); green (average economic capital, EC≥$23,200 and EC<$42,000); yellow (precarious economic capital, EC≥$10,000 and EC<$23,200); red (critical economic capital, EC<$10,000).
f. The plot shows the mean economic capital EC for active artists along the simulation.
g. The plot shows the mean symbolic capital SC gains for active artists along the simulation.

h. The plot shows the relation between mean SC for all active artists at a step (y axe) and the mean EC for all the same artists at the same step (x axe), for each step of the simulation. It is an important plot to understand if and when a self-reinforcing process between SC and EC happens.
i. The plot shows the mean self-realisation SR gains for active artists along the simulation.
j. The plot shows the number of jobseeker artists along the simulation. The red line represents job-seeker artists who are active in the cultural field, and the blue line represent exited jobseekers.
k. The plot shows the number of artists with secondary job along the simulation. The red line represents artists who a cultural secondary job, and the blue line represent artists with a non-cultural secondary job.
l. The plot shows the number of active artists in the field along the simulation. The red line represents active artists who have a secondary job, and the black line are active artists without a secondary job.
m. The plot shows the number of exited artists along the simulation. The red line represents exited artists who have a secondary job, and the black line are exited artists without a secondary job.

7. Running the model

To run the model just press the 'Default setup' button, then press 'Go (forever)' button or 'Go (by step)' button. Observe the results on the dashboard.

Tu run the model for tests, use the sliders to select parameters values, then press the 'Test Setup' button, then press 'Go (to end)' button or 'Go (step by step)' button. Observe the results on the dashboard.

8. Data output

TRACE generates individual-level simulation data in CSV format to support analysis of artist trajectories, capital accumulation, and emergent field-level patterns. Data are recorded at every simulation step for every artist, allowing researchers to reconstruct complete career trajectories over time.

The main output file, **simulation_data.csv**, contains the following variables:

Variable	Description
Tick	Simulation step (start from 0). 1 tick = 1 week.
ArtistID	Unique identifier for each artist agent (start from 0)
EC	Artist's economic capital at the end of the simulation step
SC	Artist's symbolic capital at the end of the simulation step
SR	Artist's self-realisation at the end of the simulation step
State	Artist status: Active or Exited
Colour	Visual status indicator based on economic capital thresholds

InvestmentDisposition	Artist-specific investment disposition parameter
SelfRealisationPropensity	Artist-specific self-realisation propensity parameter
Investment	Amount of economic capital invested in artistic practice during the simulation step
RecognitionEvent	Indicates whether the artist received a recognition opportunity during the simulation step (True or False)
SCGain	Symbolic capital gained from recognition during the simulation step
EconomicEvent	Indicates whether the artist received an economic opportunity during the simulation step
ECGain	Economic capital gained from recognition during the simulation step
SRGain	Self-realisation gained from artistic practice during the simulation step
Jobseeker	Binary (True/False) parameter indicating whether an artist is seeking for a job (triggered by EC<8 investments or state = exited)
SecondaryJobType	It can be Cultural or Non-cultural
SecondaryJobHours	Represents hours/week for a job (between 5-42)
SecondaryJobWage	Represents the $/hour for a job (between $50-$120)
SecondaryJobDuration	Represents the number of weeks/simulation ticks of a job (between 2-26 ticks)
JobECGain	Economic return from secondary job
ArtisticHours	Total number of hours an artist dedicate to artistic practice. Is reduced when a secondary job is taken, or it becomes 0 when an artist becomes exited from the cultural field.
ExitCount	Cumulative count of the number of times an artist exit the field (state change from active to exited)
ReturnCount	Cumulative count of the number of times an artist re-enter the field (state change from exited to active)
SecondaryJobCount	Cumulative count of the number of times an artist had a secondary job

The output file enables analysis of both individual trajectories and aggregate dynamics, including capital accumulation patterns, exit processes, opportunity distribution, and the emergence of differentiated career pathways. The recorded tick-by-tick data also support further analyses such as sequence analysis, clustering of career trajectories, and sensitivity analysis of model behaviour.

In addition to the individual-level simulation data, TRACE generates a configuration CSV file containing the complete set of model parameters used for each simulation run. This file provides a record of the experimental conditions under which the simulation was conducted and supports reproducibility, comparison between scenarios, and sensitivity analysis.

The configuration file, **model_configuration.csv**, contains:

Variable	**Description**
NumberArtists	Number of artist agents in the simulation
SimulationLength	Total number of simulation steps
RecognitionOpportunities	Number of recognition opportunities available per step
EconomicOpportunities	Number of economic opportunities available per step
SecondaryJobs	Number of secondary jobs available per step

PCulturalJobs	% of all secondary jobs which are of type cutural
InitialECmin / InitialECmax	Minimum and maximum initial EC assigned to artists
InitialSC	Initial symbolic capital assigned to artists
SC-min / SC-max	Minimum and maximum symbolic capital gain from recognition events
EC-min / EC-max	Minimum and maximum economic capital gain from economic opportunities
MinSRpropensity / MaxSRpropensity	Range used to assign artist self-realisation propensity values
MinDisposition / MaxDisposition	Range used to assign artist investment disposition values
Imin / Imax	Minimum and maximum investment parameters
K	Investment scaling parameter controlling the relationship between economic capital and investment amount.
WeekHours	Total number of working hours per week/tick.
T-cyan/ T-green/ T-yellow/ T-red	Economic capital thresholds used for visual classification of artists

Together, the simulation data output and the configuration file provide a complete record of both model dynamics and parameter settings, enabling replication of experiments and systematic comparison across different model configurations.

9. Reproducibility

The results reported in the cited academic article is based on 100 independent simulations conducted using the default parameter configuration. Each simulation uses the same model settings while allowing stochastic processes within the model to generate variation in individual trajectories and aggregate outcomes. The default parameters and samples of simulation outputs are provided to support replication and further analysis.

10. Model assumptions and limitations

Artists are initially reasonably homogeneous, with individual variations of EC, investment disposition and self-realisation propensity, and the same working hours, and initial symbolic capital and self-realisation. The cultural field is represented as closed, with no new artists entering the system along the simulation. Recognition allocation is simplified. Economic opportunities are externally generated. Institutional actors are not explicitly represented. Capital conversion mechanisms are simplified. Job allocation is simplified. Self-realisation account is simplified.

11. Future extensions:

Future developments of TRACE-v2 may extend the model to explore additional dimensions of artistic careers and cultural fields. Potential extensions include the incorporation of policy interventions, multiple interacting fields, institutional actors, alternative recognition mechanisms, demographic inequalities, and network effects.

These developments would allow further investigation of how structural conditions and institutional arrangements influence the emergence of diverse career trajectories.

12. Copyright, licensing, reuse and citation

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> *Leao, S. Z. (forthcoming, 2027) Sustaining an artistic career: a model of work, time, economic survival, and self-realisation. Creative Industries Journal.*

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Researchers who develop substantial extensions, adaptations, or new versions of TRACE are encouraged to notify the author(s). This supports documentation of model evolution, ensures appropriate acknowledgement of the original model, and facilitates potential collaboration within the research community.

13. Contact and collaboration

TRACE-v2 is part of an ongoing research program in computational sociology and digital humanities. The author welcomes the dissemination, reuse, and further development of the model and are interested in contributing to collaborative research exploring the application of computational approaches to the study of artistic careers and cultural fields.

For questions about TRACE-v2, clarification regarding model use, or discussion of potential collaborations, please contact:

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